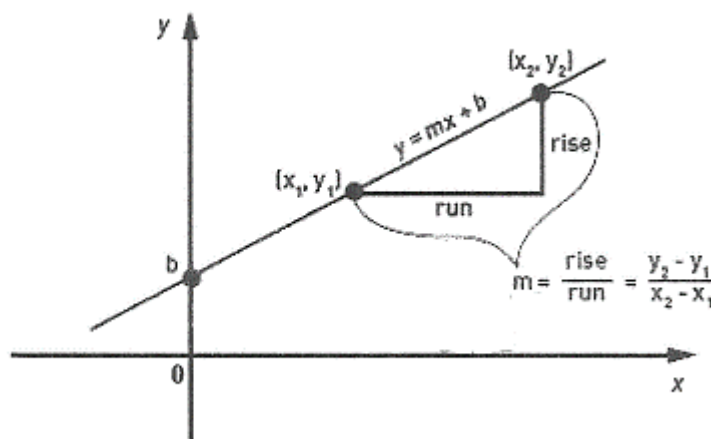


Slope

If the graph plotted on y against x is a straight line graph, the relation between the variable x and y is said to be linear relation. The relationship between x and y can be expressed in the form of $y = mx + b$ where m and b are constants (see figure 1)

Figure 1

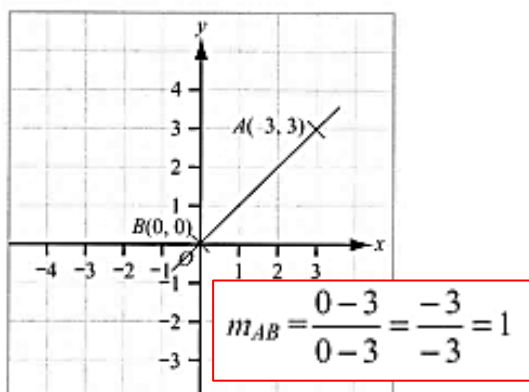


$$(\text{Slope}) \quad m = \frac{y_2 - y_1}{x_2 - x_1}$$

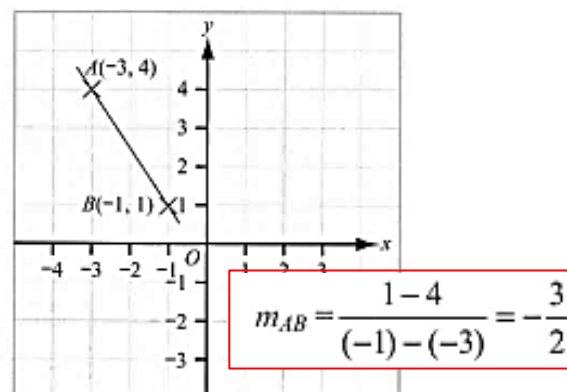
$b = y - \text{intercept}$

Find the slope m_{AB} of the straight line AB in each of the following.

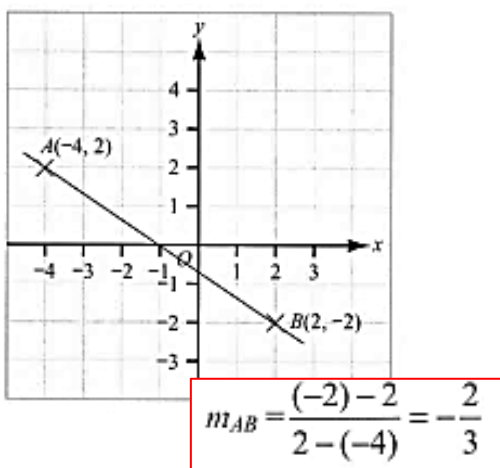
1.



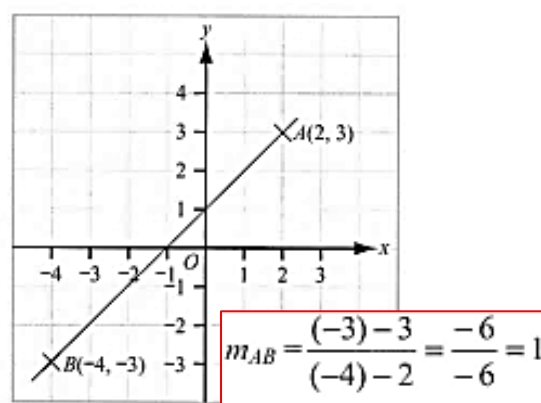
2.



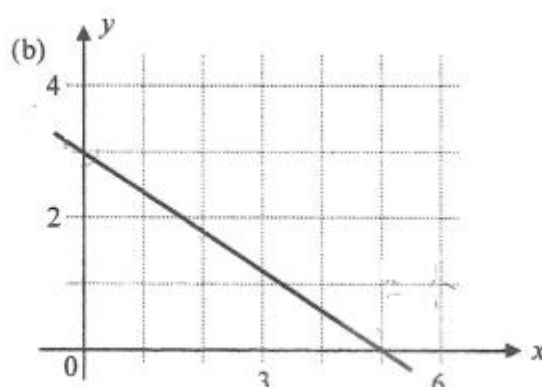
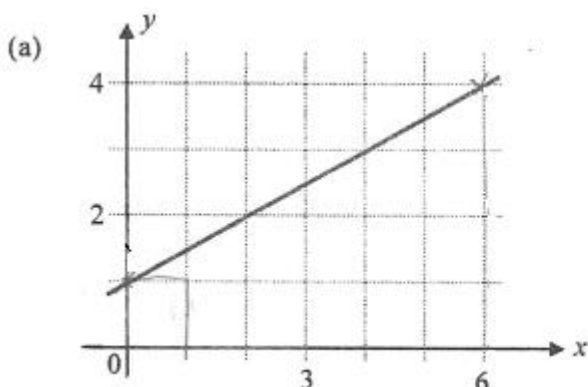
3.



4.



5. Find the slope and the y-intercept of the following straight line graphs. Hence write down the equation of the straight line



Answer: Slope = $\frac{4-1}{6-0} = \frac{3}{6} = \frac{1}{2}$

y-intercept = 1

Equation: $y = \frac{1}{2}x + 1$

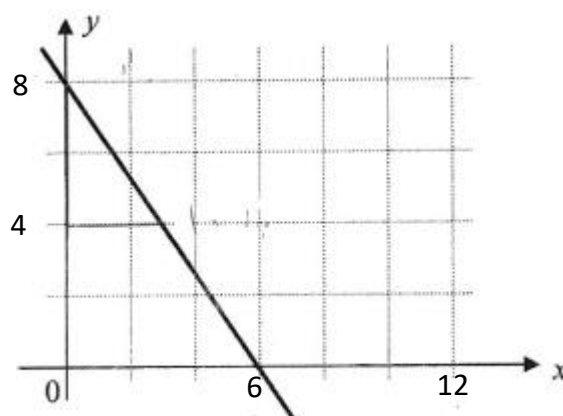
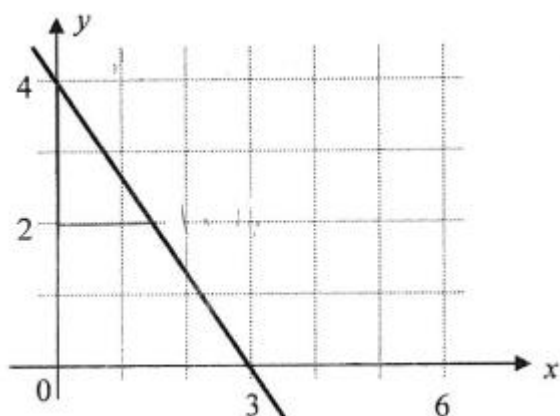
Equation = Slope + y-intercept

Answer: Slope = $\frac{3-0}{0-5} = -\frac{3}{5}$

y-intercept = 3

Equation: $y = -\frac{3}{5}x + 3$

6. Find the slope and the y-intercept of the following straight line graphs. Hence write down the equation of the straight line



Slope = $\frac{4-0}{0-3} = -\frac{4}{3}$

y-intercept = 4

Equation: $y = -\frac{4}{3}x + 4$

Equation: $y = -\frac{4}{3}x + 4$

Slope = $\frac{8-0}{0-6} = -1.33$

y-intercept = 8

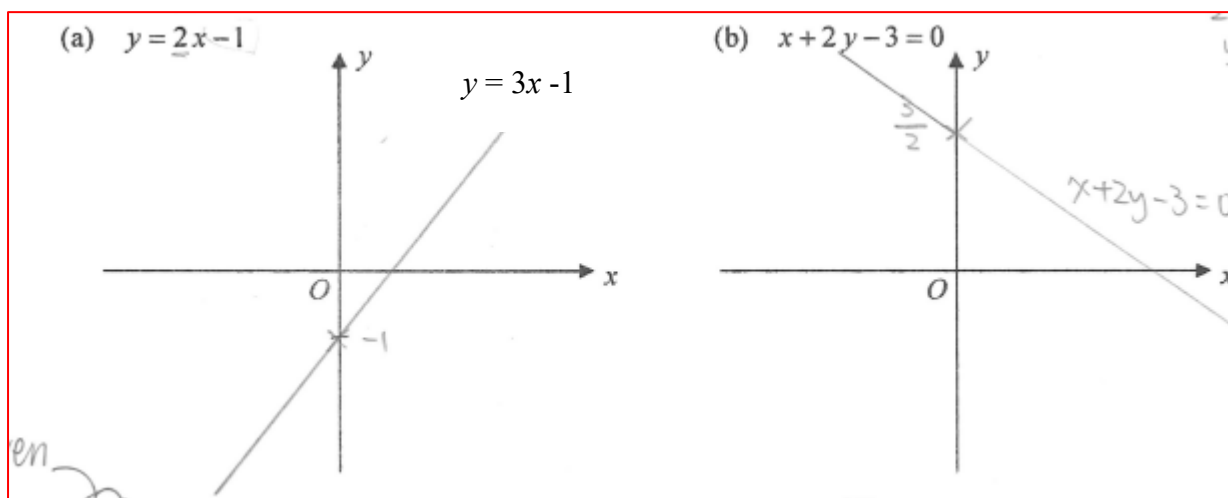
Equation: $y = -1.33x + 8$

Equation: $y = -1\frac{1}{3}x + 8$

7. Sketch the following straight line graphs:

(a) $y = 3x - 1$

(b) $x + 2y - 3 = 0$



8. The equation of a straight line is given as $8x - 5y + 40 = 0$

(a) Re-write the equation in the form of $y = mx + b$

(b) What is the slope and the y-intercept of the equation?

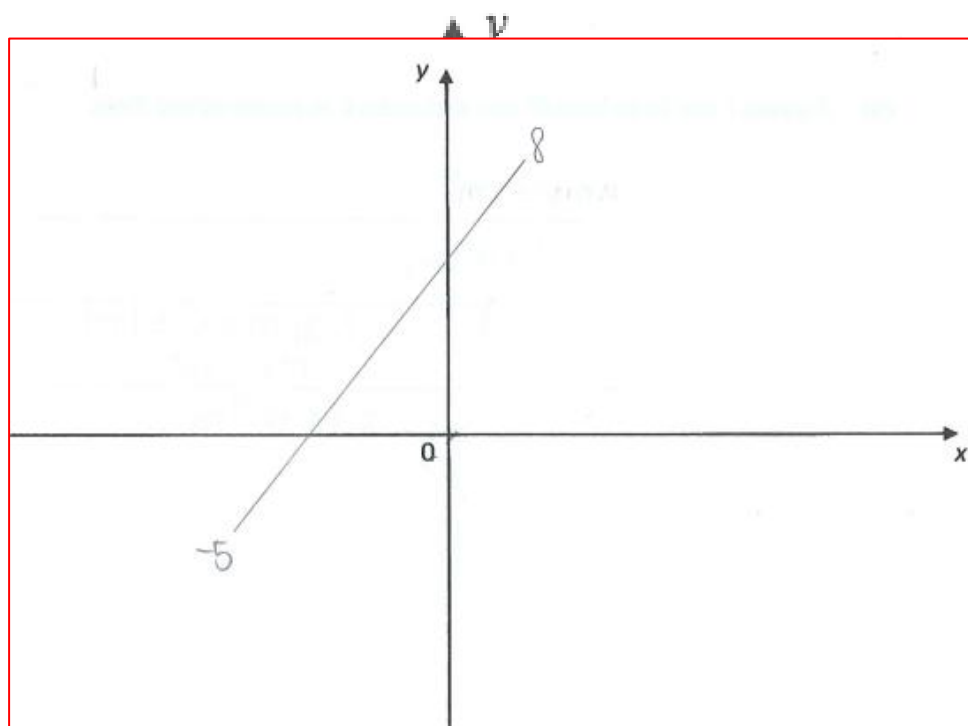
(c) Hence sketch the straight line for the equation. Indicate the x- and y- intercept clearly on the sketch

$$5y = 8x + 40$$

$$y = \frac{8}{5}x + 8$$

The slope = $\frac{8}{5}$

y intercept : 8



9. The equation of a straight line is given as $3x - 5y + 10 = 0$

(a) Re-write the equation in the form of $y = mx + b$

(b) What is the slope and the y-intercept of the equation?

(c) Hence sketch the straight line for the equation. Indicate the x- and y- intercept clearly on the sketch

Equation : $y = \frac{3}{5}x + 2$

Slope = 0.6

y-intercept = 2

